



PIMS PRO

Pipeline Integrity Management System

Pipeline Corrosion Assessment & Remaining Life Calculator

Version	1.0
Developer	M. Fahad Zafar
Report File	PL-004_Inspection_Report_2026-07-16.pdf
Generation Date	16 July 2026

Inspection Information

Pipe ID	PL-004
Pipeline Name	PL-004
Material	Carbon Steel
Fluid	Crude Oil
Inspector	Hania
Inspection Date	2026-07-16
Operating Pressure	1.0 MPa

Engineering Calculations

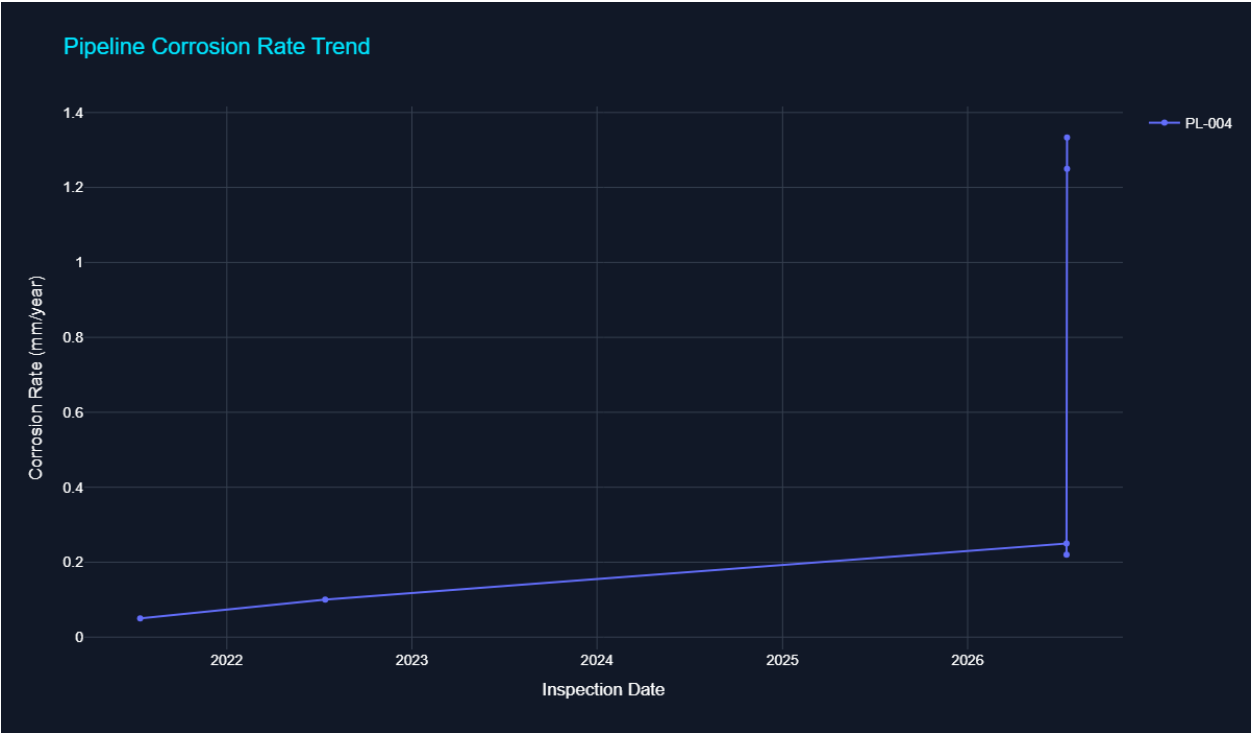
Parameter	Value
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Corrosion Rate	1.25 mm/year
Minimum Thickness	9.84 mm
Remaining Life	88.13 years
Safety Factor	12.19
Corrosion Loss	4.00 %
Health Score	78.00 %
Failure Year	2114

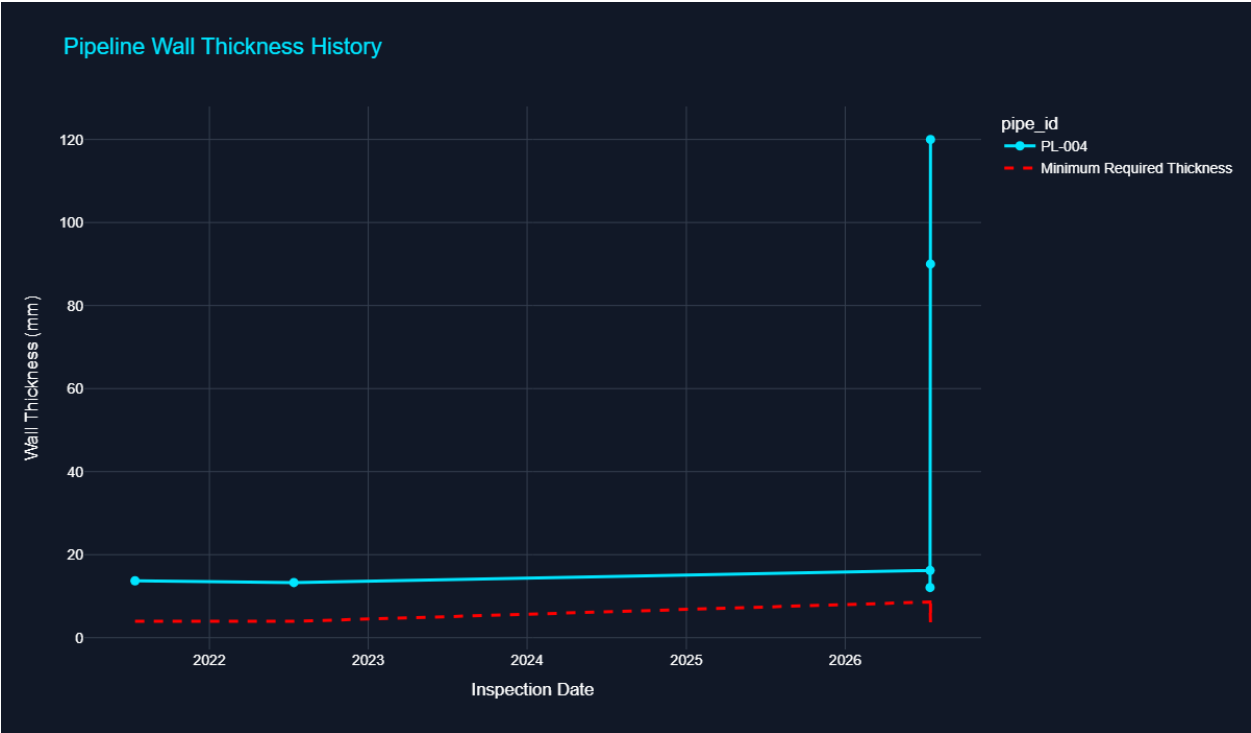
KPI Summary

Health Score	78.00%
Risk Level	Moderate
Remaining Life	88.13 Years
Corrosion Rate	1.25 mm/year

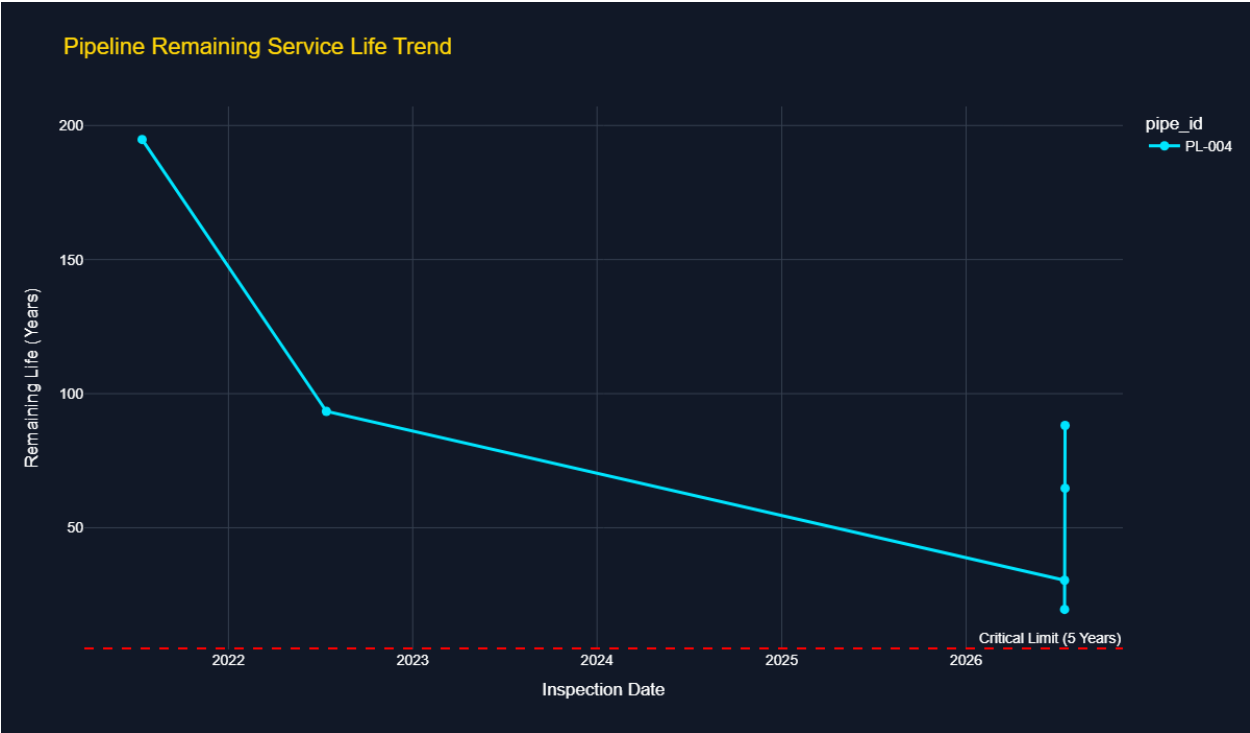
Corrosion Trend



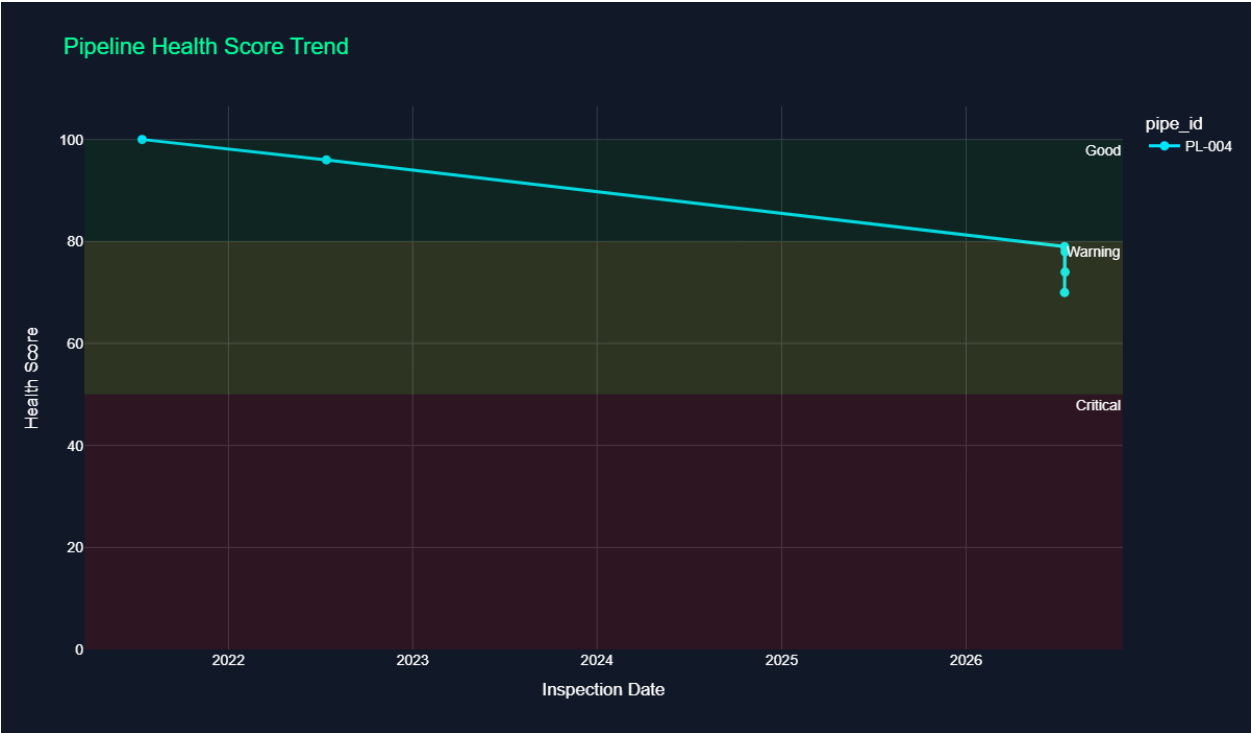
Thickness History



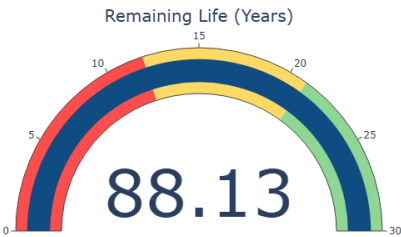
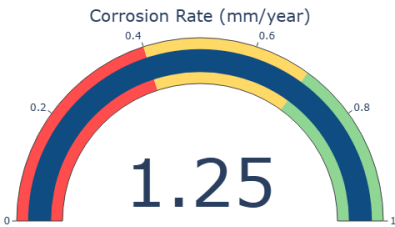
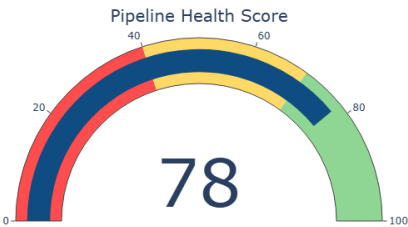
Remaining Life Trend



Health Trend



Engineering KPI Gauges

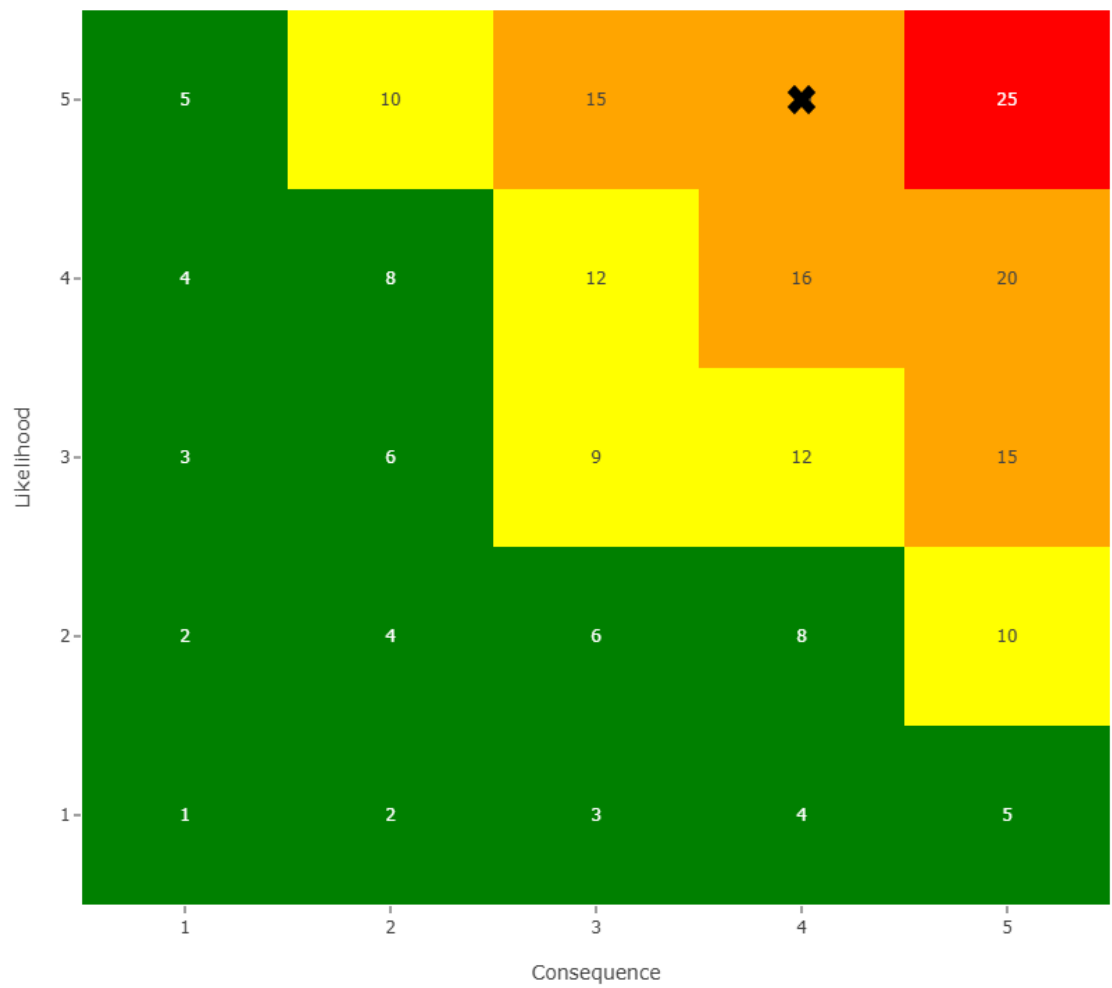


Risk Assessment

Parameter	Value
Likelihood	5
Consequence	4
Risk Score	20
Risk Category	Moderate

5 × 5 Risk Matrix

5×5 Pipeline Risk Matrix



Engineering Recommendation

Overall Decision	Pipeline Acceptable with Monitoring
Maintenance Suggestion	Increase inspection frequency.
Inspection Interval	Next inspection after 2 years.
Priority	MEDIUM
Warning Message	Moderate corrosion detected.

Inspection Summary

The pipeline (PL-004) remains suitable for continued operation. The measured corrosion rate is 1.25 mm/year with an estimated remaining service life of 88.13 years. The calculated health score is 78.0% and the overall risk category is Moderate. Increase inspection frequency.

Conclusion:

Based on the inspection data and engineering calculations performed in accordance with ASME B31.3 and API 570 principles, the pipeline integrity has been evaluated and appropriate maintenance recommendations have been provided to support safe and reliable operation.

Engineering Notes

Engineer observations, maintenance remarks, or site comments may be recorded below.

Engineering Standards

- ASME B31.3 – Process Piping

- API 570 – Piping Inspection Code

The engineering calculations, integrity assessment, corrosion evaluation, remaining life estimation, and inspection recommendations presented in this report are based on the principles and methodologies defined in ASME B31.3 and API 570. The report is intended as an engineering decision-support document and should be reviewed by qualified personnel before maintenance or operational decisions are made.